Nishi Kanta Paul

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Research Profile

Medical AI researcher developing efficient, explainable deep learning for medical imaging and biosignals, with a focus on lightweight architectures, segmentation, and rigorous evaluation. Founder of NIMISHES Lab and author of **4 peer-reviewed papers** published or accepted at IEEE and international conferences.

Research Interests

Efficient AI · Medical Image Analysis · Vision-Language Models · Trustworthy & Explainable AI · Biosignals

Education

B.Sc. in Computer Science and Engineering *2020 – 2024*
Ahsanullah University of Science and Technology (AUST)
CGPA: **3.749 / 4.00** • Ranked **7th / 100 (Top 7%)**

Research Experience

Founder & Research Lead — *NIMISHES Lab* *Apr 2026 – Present*

- Collaborate with researchers at nine institutions in the USA, UAE, France, and Austria.
- First-authored lightweight models for EEG sleep staging (83.9% on Sleep-EDF, **3–5× fewer parameters**) [1] and retinopathy grading (QWK 0.884 on APTOS 2019, beating ResNet50 at a fifth of its size) [3].
- First-authored fMRI-privileged, EEG-only decoding via optimal transport (0.801 ROC-AUC) [9] and chest X-ray anonymization (re-identification AUC 0.961 → 0.791 for a 0.012 AUROC drop) [10].
- Co-developed an 11.1M-parameter image-only chest X-ray grounding model (F1 0.302 → 0.338) [8] and led agentic medical vision work [11], supervised by Dr. Jamil Fayyad (American University of Sharjah).

Remote Research Assistant — *Washington State University* *Jun 2026 – Present*

Project: Anomaly-Aware ForensiBlock *(supervised by Prof. Asma Jodeiri Akbarfam)*

- Contributing to the ML and explainable AI pipeline for real-time user anomaly detection.
- Developing anomaly detection and XAI integration for traceable forensic decisions; co-author of [6].

ELITE Research Lab *Dec 2025 – Present*

Research Assistant *Sep 2026 – Present*

Project: LCM-UNet — Reparameterizable Local-Compensated Mamba U-Net *(Joint First Author, [5])*

- Developed the Residual Dilated Local Adapter (RDLA): sharper boundaries, no inference overhead.
- Improved IoU by +0.48 / +0.37 pp on ISIC 2017 / 2018 and reached $86.37 \pm 0.29\%$ IoU on PH^2 with only **34,180 parameters and 0.06 GFLOPs**, under multi-seed and statistical evaluation.

Junior Research Assistant *Mar 2026 – Aug 2026*

Project: BGD-SF PolySegNet — Boundary-Guided Polyp Segmentation *(First Author, [7])*

- Developed boundary-guided multi-scale aggregation, selective skip fusion, and boundary refinement; trained on 3,535 images from three datasets and tested on four external ones.
- Achieved **0.887 Dice and 0.839 IoU**, with +15.7% Boundary F1 and −27.5% HD95 / −34.6% ASSD versus the strongest CNN baselines.

Research Intern *Dec 2025 – Feb 2026*

- Supported medical imaging research through data preprocessing, baseline reproduction, and experiments.

Publications

Published & Accepted

[1] **N. K. Paul**, M. S. I. Shovo, I. J. Esha, and A. Rahman, "SleepEffFormer: Efficient CNN-Transformer with Transition-Aware Smoothing for Single-Channel EEG Sleep Stage Classification," in *Proc. 5th IEEE Int. Conf. on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2026. [Paper] [arXiv] [Code]

[2] M. S. I. Shovo, **N. K. Paul**, A. Rahman, and I. J. Esha, "WaveFoG: Wavelet-Gated Transformer for Parkinson's Freezing of Gait Detection Using Wearable Accelerometer Signals," in *Proc. 5th IEEE BECITHCON*, Dhaka, Bangladesh, 2026. [Paper] [arXiv] [Code]

[3] **N. K. Paul**, M. S. I. Shovo, A. Chowdhury, and I. J. Esha, "DR-LiteNet: A Lightweight Explainable Hybrid CNN Framework for Imbalanced Diabetic Retinopathy Grading via Adaptive SMOTE Fusion," in *Proc. 1st Int. Conf. on Next-Generation Electrical & Electronics, Computer Systems, and Technologies (iCONEECT)*, Chittagong, Bangladesh, 2026. [Paper] [Code]

[4] M. M. Kabir, M. A. Nawar, M. I. Remon, **N. K. Paul**, H. Bhuiyan, and M. M. Hoque, "Video-based Driver Distraction Detection using Transfer Learning: Bangladesh Perspective," in *Proc. Int. Conf. on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)*, 2024. [IEEE Xplore]

Under Review

[5] M. S. I. Shovo*, **N. K. Paul***, and K. Morol, "LCM-UNet: A Reparameterizable Local-Compensated Mamba U-Net for Skin Lesion Segmentation," *IEEE Journal of Biomedical and Health Informatics*, under review. (*Equal contribution.)

[6] A. Jodeiri Akbarfam, **N. K. Paul**, and S. Ismail, "Anomaly-Aware ForensiBlock: Explainable Behavioral Monitoring for Digital Evidence Access," *IEEE Access*, under review.

[7] **N. K. Paul**, S. Maherin, and M. Akter, "BGD-SF PolySegNet: Boundary-Guided Dynamic Selective Fusion Network for Robust Polyp Segmentation in Colonoscopy Images," *PLOS ONE*, under review.

[8] M. S. I. Shovo, **N. K. Paul**, and J. Fayyad, "APED-CXR: Adaptive Privileged Evidence Distillation for Chest X-Ray Grounding with Image-Only Inference," *IEEE Int. Symp. on Biomedical Imaging (ISBI)* 2027, under review. [arXiv]

[9] **N. K. Paul**, M. S. I. Shovo, and D. Mahapatra, "Band2Net: Frequency-to-Network Transport Distillation for fMRI-Privileged, EEG-Only Session-Condition Decoding," *IEEE ISBI* 2027, under review.

[10] **N. K. Paul**, M. S. I. Shovo, and D. Sidibé, "LAPIS: Clinically Constrained Patient-Identity Residual Learning for Chest X-Ray Anonymization," *IEEE ISBI* 2027, under review.

[11] **N. K. Paul**, M. S. I. Shovo, J. Fayyad, and M. M. Rahman, "Coupled or Independent? Fault Coupling in Agentic Medical Vision Systems," *Workshop on Foundation AI for Biomedical Reasoning, Imaging, and Cognition (FABRIC), IEEE/CVF WACV* 2027, under review.

[12] M. S. I. Shovo, **N. K. Paul**, T. T. Dong, and X. Lin, "AquaProbe: Efficient Adaptation of Frozen Vision Models for Underwater Images," *5th Workshop on Maritime Computer Vision (MaCVi), IEEE/CVF WACV* 2027, under review.

In Preparation

[13] **N. K. Paul**, M. S. I. Shovo, and C. González, "CoMAF-Polyp: Calibrated Reliability-Aware Fusion of Specialist and Foundation Pseudo-Labels for Semi-Supervised Polyp Segmentation," target venue: *Medical Imaging with Deep Learning (MIDL)* 2027.

Industry Experience

Software Engineer I — *Brain Station 23 PLC* *Mar 2025 – Present*

- **OneID** (FastAPI, PostgreSQL, Keycloak): built an eKYC pipeline that cut integration latency by **40%** and a JSONB metadata system that sped up document processing by 35%.
- Added NID validation and normalization that cut invalid submissions by 30%, and secured sensitive identity data with Keycloak.
- **ObhaiVTS** (React, TanStack Query, Zustand): delivered invoice, customer, vehicle, and hub management flows; server-side pagination and dynamic filtering improved data handling by 30%.

Teaching Experience

Adjunct Lecturer, CSE Department — *International Standard University* *Jan 2026 – Present*

- Teaching Robotics, IoT, and Algorithms to about 60 undergraduates and designing hands-on labs.

Competitive Programming Mentor (Remote) — *Shohoj Coding* *Sep 2024 – Dec 2025*

- Mentored 50+ students in graph theory, number theory, and data structures.

Selected Academic Honors

Scholarship for Outstanding Academic Performance (GPA 3.98 / 4.00, ranked 1st / 100) *Fall 2020*
Khondakar Ahmed Muktadir & Mosammat Khodeza Begum Scholarship *Fall 2022*

Technical Skills

Deep Learning Modeling	PyTorch, TensorFlow, Transformers, CLIP, LLaVA, BioViL, Grad-CAM, SHAP CNNs, Vision Transformers, Mamba / state-space models, multimodal fusion, knowledge distillation, optimal transport, calibration, structural reparameterization
Data & Systems	FastAPI, Django, PostgreSQL, MongoDB, Keycloak, Docker, Linux, React
Programming	Python, C++, JavaScript, TypeScript

Additional

Competitive Programming: ICPC Asia Dhaka Regional 2023 (116th) · Intra-AUST 2022 Champion · Codeforces Specialist (max rating 1421, 1000+ problems) · CodeChef (max rating 1570) · LeetCode (500+ problems).

Problem Setter & Judge: AUST CSE Carnival programming contests and Intra-AUST contests (2023 – 2025).

References

Prof. Asma Jodeiri Akbarfam
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Research supervisor, Anomaly-Aware ForensiBlock

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